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The Polynomial Method and the Endpoint Case of the Multilinear Keakeya Estimate

Author:

Seunghyun Lee (CID: 02411618)

Akshay Popat (CID: 02204564)

Ziyu Wang (CID: 02395271)

James Wenham (CID: 02386619)

Shengyu Zhang (CID: 02419829)

Supervisor(s):

Dr Zhiyuan Zhang

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Abstract

We look at the endpoint case of the multilinear Kakeya estimate, first proved by Guth. Our goal is to understand the estimate with the Loomis-Whitney inequality, illustrate Guth's polynomial method of proof in detail and suggest further applications of the estimate. We also discuss the tools used in the proof: the Borsuk-Ulam theorem, the polynomial ham sandwich theorem and the cylinder estimate.

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1 Introduction

In the 1970s, Stein introduced the *restriction problem*, a conjecture that remains unsolved in Fourier analysis. Since then, the restriction problem has been shown to be related to various problems in Fourier analysis, dynamical systems, combinatorics and geometric measure theory (as will be discussed in this paper). Hence, solving the restriction problem is one of the major goals of modern Fourier analysis.

We begin this discussion by presenting the formal statement of the restriction problem. Let $f : \mathbb{R}^n \rightarrow \mathbb{C}$ be some function in $L^p(\mathbb{R}^n)$, and define the Fourier transform of f as

$$\hat{f}(\xi) = \int_{\mathbb{R}^n} f(x) e^{-ix \cdot \xi} dx,$$

the inverse Fourier transform as

$$f(x) = \int_{\mathbb{R}^n} \hat{f}(\xi) e^{ix \cdot \xi} d\xi.$$

The restriction problem concerns the restriction of $\hat{f}$ to some subset S of $\mathbb{R}^n$ and asks whether there exists some constant bound $C_{p,q,n,S}$ such that

$$\left\| \hat{f}|_S \right\|_{L^q(S)} \leq C_{p,q,n,S} \|f\|_{L^p(\mathbb{R}^n)}, \quad (1)$$

where the notation $C_{p,q,n,S}$ is used to indicate that the constant depends only on p, q, n, S .

Another formulation of this problem is as follows:

Conjecture 1.1 (Linear restriction problem, [1]). *Let $n \geq 2$, U be a compact neighbourhood of the origin in $\mathbb{R}^{n-1}$ and $\Sigma : U \rightarrow \mathbb{R}^n$ be a smooth parametrisation of a $(n-1)$ -dimensional submanifold S of $\mathbb{R}^n$. For a given Σ , we define the extension operator $\mathcal{E}$ to be*

$$\mathcal{E}g(\xi) = \int_U g(x) e^{i\xi \cdot \Sigma(x)} dx, \quad (2)$$

where $g \in L^1(U)$ and $\xi \in \mathbb{R}^n$. (Notice that the adjoint of this operator $\mathcal{E}^*$ satisfies $\mathcal{E}^*f = \hat{f} \circ \Sigma$, implying that $\mathcal{E}^*f$ is the restriction of $\hat{f}$ to $S \subset \mathbb{R}^n$. Hence, $\mathcal{E}$ is also called the adjoint restriction operator.)

If Σ has everywhere non-vanishing Gaussian curvature, $q > \frac{2n}{n-1}$ and $p' \leq \frac{n-1}{n+1}q$ (where p' is the conjugate exponent of p), then there exists a constant $C_{n,\Sigma}$ such that

$$\|\mathcal{E}g\|_{L^q(\mathbb{R}^d)} \leq C_{n,\Sigma} \|g\|_{L^p(U)}$$

for all $g \in L^p(U)$.

Although it is still a conjecture, there has been successful progress in partially solving this problem, such as the Tomas-Stein restriction theorem and the (linear) Kakeya estimate. These theorems have provided the existence of a constant bound in specific cases where the exponents p, q have been restricted to a certain value, or the function f holds specific conditions. Here, we are interested in the Kakeya estimate.

The Kakeya estimate arose from a different problem, called the *Kakeya set conjecture*. The Kakeya set conjecture is a problem in geometric measure theory that also remains unsolved,

which concerns the properties of a set in $\mathbb{R}^n$ that contains a unit line segment in every direction (also known as a *Keakeya set*).

Later, it was shown that the linear restriction problem implies the Keakeya set conjecture [1]. This intimate connection can be briefly observed in the alternate, stronger version of the Keakeya set conjecture, known as the *(linear) Keakeya estimate*. It can be proved that the linear Keakeya estimate implies the Keakeya set conjecture [4].

Conjecture 1.2 (Linear Keakeya estimate, [1]). *Let $T_1, \dots, T_A$ be arbitrary cylinders in $\mathbb{R}^n$ with radius $\delta \in (0, 1)$ and length 1 such that their orientations form a δ -separated set of points on S^{n-1} . If $\frac{n}{n-1} < q \leq \infty$, then there exists constant C_n such that*

$$\left\| \sum_{a=1}^A 1_{T_a} \right\|_{L^q(\mathbb{R}^n)} \leq C_n \delta^{(n-1)/q} A^{1/q}.$$

Research on these problems has led to their multilinear analogues. Below, we state the multilinear restriction problem and the multilinear Keakeya estimate.

Conjecture 1.3 (Multilinear restriction problem, [1]). *For $1 \leq j \leq n$, let U_j be some compact set in $\mathbb{R}^{n-1}$ and $\Sigma_j : U_j \rightarrow \mathbb{R}^n$ be a smooth mapping with corresponding extension operator $\mathcal{E}_j$ as in (2). Under certain geometric conditions depending on constants $A, \nu > 0$:*

for $q \geq \frac{2n}{n-1}$ and $p' \leq \frac{n-1}{n}q$, there exists constant $C_{A,\nu,n,U}$ such that

$$\left\| \prod_{j=1}^n \mathcal{E}_j g_j \right\|_{L^{q/n}(\mathbb{R}^n)} \leq C_{A,\nu,n,U} \prod_{j=1}^n \|g_j\|_{L^p(U_j)}$$

for all $g_1 \in L^p(U_1), \dots, g_n \in L^p(U_n)$, where U denotes the collection of sets $U_1, \dots, U_n$.

Theorem 1.4 (Multilinear Keakeya estimate). *Let $T_{j,a}$ be arbitrary cylinders in $\mathbb{R}^n$ with radius $\delta \in (0, 1)$ and length 1, ranging over $1 \leq j \leq n$ and $1 \leq a \leq A(j)$. Suppose $T_{j,a}$ is parallel to some direction within a sufficiently small fixed neighbourhood θ of the x_j -axis (i.e., the collections are transversal). If $\frac{n}{n-1} \leq q \leq \infty$, then there exists constant $C_{n,\theta}$ such that*

$$\left\| \prod_{j=1}^n \left(\sum_{a=1}^{A(j)} 1_{T_{j,a}} \right) \right\|_{L^{q/n}(\mathbb{R}^n)} \leq C_{n,\theta} \prod_{j=1}^n \delta^{n/q} A(j). \quad (3)$$

In [1], Bennett, Carbery and Tao successfully proved the multilinear Keakeya estimate for $\frac{n}{n-1} < q \leq \infty$ using the heat flow method. The endpoint case of the multilinear Keakeya estimate, when $q = \frac{n}{n-1}$, was later proved by Guth in [6] using the polynomial method. Guth's statement of the endpoint case works under a weaker transversality condition: if $v_{j,a}$ is a unit vector parallel to the cylinder $T_{j,a}$, then the determinant of the matrix $(v_{1,a_1}, \dots, v_{n,a_n})$ has a norm of at least $\theta > 0$ where $a_j \in \{1, \dots, A(j)\}$.

Substituting the endpoint case $q = \frac{n}{n-1}$ into (3):

$$\left\| \prod_{j=1}^n \left(\sum_{a=1}^{A(j)} 1_{T_{j,a}} \right) \right\|_{L^{\frac{1}{n-1}}(\mathbb{R}^n)} \leq C_{n,\theta} \prod_{j=1}^n \delta^{n-1} A(j).$$

Expressing the norm as an integral, we get

$$\int_{\mathbb{R}^n} \prod_{j=1}^n \left(\sum_{a=1}^{A(j)} 1_{T_{j,a}} \right)^{\frac{1}{n-1}} \leq C_{n,\theta} \prod_{j=1}^n \delta A(j)^{\frac{1}{n-1}} \quad (4)$$

and assuming that the $T_{j,a}$ are cylinders with radius 1 and infinite length, the inequality becomes

$$\int_{\mathbb{R}^n} \prod_{j=1}^n \left(\sum_{a=1}^{A(j)} 1_{T_{j,a}} \right)^{\frac{1}{n-1}} \leq C_{n,\theta} \prod_{j=1}^n A(j)^{\frac{1}{n-1}}. \quad (5)$$

This is the form of the endpoint case of the multilinear Kakeya estimate introduced and proved in [6].

In this paper, we are interested in a weaker version of the above inequality. This theorem was also introduced and proved by Guth in [6]:

Theorem 1.5 (Weak endpoint case of the multilinear Kakeya estimate, [6]). *Let $T_{j,a}$ be arbitrary cylinders in $\mathbb{R}^n$ with radius 1 and infinite length, ranging over $1 \leq j \leq n$ and $1 \leq a \leq A$. If $T_{j,a}$ are parallel to some direction within an angle of $(100n)^{-1}$ to the x_j -axis and*

$$I = \bigcap_{j=1}^n \bigcup_{a=1}^A T_{j,a},$$

then there exists some constant C_n such that $\text{Vol}(I) \leq C_n A^{\frac{n}{n-1}}$.

By the properties of the indicator function, we see that

$$\text{Vol}(I) = \int_{\mathbb{R}^n} (1_I)^{\frac{1}{n-1}} \leq \int_{\mathbb{R}^n} \prod_{j=1}^n \left(\sum_{a=1}^{A(j)} 1_{T_{j,a}} \right)^{\frac{1}{n-1}}, \quad (6)$$

which implies that 1.5 is consistent with (5) and further, Theorem 1.4.

We are interested in Theorem 1.5 for multiple reasons. First, the proof of Theorem 1.5 presented in [6] uses the polynomial ham sandwich theorem, which is a distinct approach from known proofs of other versions of the Kakeya estimates. The same method is eventually adapted for proving the endpoint case of the multilinear Kakeya estimate. We will introduce the polynomial ham sandwich theorem in Section 4 and its lemma, the Borsuk-Ulam theorem, in Section 3.

Second, the constant C_n in Theorem 1.5 is independent of θ , unlike in Theorem 1.4 or (5). This follows from the special geometric inequalities that the angle value $(100n)^{-1}$ gives. We observe this phenomenon and prove the so-called cylinder estimate in Section 5. Section 6 will cover the proof of Theorem 1.5.

Lastly, the endpoint case of the multilinear Kakeya estimate leads to other significant results. As we will discuss in Section 7, using results from complex analysis, the endpoint case leads to the case of all exponents of the multilinear Kakeya estimate.

The main objective of this paper is to go through the proof of Theorem 1.5 in full detail and give the intuitions behind the tools used in the proof. We begin this discussion in the next section by looking at a geometric inequality closely related to the endpoint case of the multilinear Kakeya estimate.

2 Loomis-Whitney inequality

In this section, we aim to give a geometric illustration of Theorem 1.5 using the Loomis-Whitney inequality. The Loomis-Whitney inequality was first introduced by Loomis and Whitney in [9]. It is a geometric inequality motivated by the isoperimetric inequality.

2.1 Proof of the Loomis-Whitney inequality

We begin by introducing and proving the Loomis-Whitney inequality. It comes with two versions: the original version introduced by Loomis and Whitney in [9] and the general version, which extends the original version to an inequality for arbitrary functions in the L^1 -space.

Both versions of the inequality require Hölder's inequality in their proofs, so we quickly mention these:

Theorem 2.1 (Hölder's inequality). *Let $p, q \in [1, \infty]$ satisfy $1/p + 1/q = 1$. If $f \in L^p(\mathbb{R}^n)$ and $g \in L^q(\mathbb{R}^n)$, then $fg \in L^1(\mathbb{R}^n)$ and*

$$\|fg\|_{L^1(\mathbb{R}^n)} \leq \|f\|_{L^p(\mathbb{R}^n)} \|g\|_{L^q(\mathbb{R}^n)}.$$

Theorem 2.2 (Hölder's inequality, general ver.). *Let $p_1, \dots, p_n, q \in [1, \infty]$ satisfy $1/p_1 + \dots + 1/p_n = 1/q$. If $f_1, \dots, f_n$ are functions in $L^{p_1}(\mathbb{R}^n), \dots, L^{p_n}(\mathbb{R}^n)$, respectively, then*

$$\left\| \prod_{j=1}^n f_j \right\|_{L^q(\mathbb{R}^n)} \leq \prod_{j=1}^n \|f_j\|_{L^{p_j}(\mathbb{R}^n)}.$$

Return to the original version of the Loomis-Whitney inequality.

Theorem 2.3 (Loomis-Whitney inequality, original ver.). *Let E be an open set in $\mathbb{R}^n$. Define $\pi_j : \mathbb{R}^n \rightarrow \mathbb{R}^{n-1}$ to be the projection map onto the j th coordinate hyperplane, i.e., $\pi_j(x_1, \dots, x_n) = (x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_n)$. Then,*

$$\text{Vol}(E) \leq \left(\prod_{j=1}^n \text{Vol}(\pi_j(E)) \right)^{\frac{1}{n-1}}.$$

We shall look at the proof given in [9]. The idea is to consider a set of unit cubes in the integer lattice grid of $\mathbb{R}^n$ first. In other words, we are considering a set of unit cubes with vertices that have integer-valued coordinates.

Lemma 2.4. *Let S be a set of unit cubes in the integer lattice grid of $\mathbb{R}^n$ and S_i be the set of cubes obtained by projecting cubes in S onto the i th coordinate hyperplane. If N is the number of cubes in S and N_i is the number of cubes in S_i , then*

$$N \leq \left(\prod_{j=1}^n N_j \right)^{\frac{1}{n-1}}.$$

Proof. Apply induction on n . The case where $n = 2$ is straightforward, since $N_1 N_2$ is equal to the maximum number of unit cubes that can fit inside a rectangle that includes all cubes in S .

Now assume that the inequality holds for $n - 1$. Notice that projecting each unit cube in S onto the first coordinate axis produces a unit interval in $\mathbb{R}$ with integer endpoints. Let $I_1, \dots, I_k$ be the unit intervals obtained by projecting the cubes in S . Also, define T_i to be the set of cubes projecting onto I_i , and T_{ij} be the set of cubes obtained by projecting cubes in T_i onto the j th coordinate hyperplane (i.e., $T_{ij} = \pi_j(T_i)$).

Let a_i be the number of cubes in T_i and a_{ij} be the number of cubes in T_{ij} . Since the disjoint union of T_i 's is equal to S ,

$$\sum_{i=1}^k a_i = N$$

and $a_i \leq N_1$. Similarly,

$$\sum_{i=1}^k a_{ij} = N_j.$$

By applying the induction hypothesis to each T_i , we have $a_i^{n-2} \leq a_{i2} \cdots a_{in}$ for all $i = 1, \dots, k$. Next, we apply this to obtain:

$$N = \sum_{i=1}^k a_i \leq \sum_{i=1}^k \left(N_1^{\frac{1}{n-1}} \prod_{j=2}^n a_{ij}^{\frac{1}{n-1}} \right) = N_1^{\frac{1}{n-1}} \sum_{i=1}^k \prod_{j=2}^n a_{ij}^{\frac{1}{n-1}}.$$

Now we use the Hölder's inequality for the counting measure:

$$\sum_{i=1}^k \prod_{j=2}^n a_{ij}^{\frac{1}{n-1}} = \left\| \prod_{j=2}^n a_{ij}^{\frac{1}{n-1}} \right\|_1 \leq \prod_{j=2}^n \left\| a_{ij}^{\frac{1}{n-1}} \right\|_{n-1} = \prod_{j=2}^n \left(\sum_{i=1}^k a_{ij} \right)^{\frac{1}{n-1}}.$$

Therefore,

$$N \leq N_1^{\frac{1}{n-1}} \sum_{i=1}^k \prod_{j=2}^n a_{ij}^{\frac{1}{n-1}} \leq N_1^{\frac{1}{n-1}} \prod_{j=2}^n \left(\sum_{i=1}^k a_{ij} \right)^{\frac{1}{n-1}} = N_1^{\frac{1}{n-1}} \prod_{j=2}^n N_j^{\frac{1}{n-1}} = \prod_{j=1}^n N_j^{\frac{1}{n-1}}.$$

□

It can easily be seen that translating and scaling the set of unit cubes in the above lemma does not affect the result itself. Hence, we can apply the lemma to cases where we have cubes on some arbitrary lattice grid that partitions $\mathbb{R}^n$ into equally sized cubes.

To prove Theorem 2.3, we will approximate the given open set E in $\mathbb{R}^n$ with cubes on some lattice grid, and apply the above lemma to reach our desired result.

Proof of Theorem 2.3. Let $\delta > 0$ and S be a set of cubes on the lattice grid of $\mathbb{R}^n$ with side length δ . For arbitrary $\epsilon > 0$, choose δ small enough such that $\text{Vol}(E \setminus \bar{S}) < \epsilon$, where $\bar{S}$ is the closure of S . Then, applying Lemma 2.4,

$$(\text{Vol}(\bar{S}))^{n-1} = (N\delta^n)^{n-1} = N^{n-1} \delta^{n(n-1)} \leq \prod_{j=1}^n N_j \delta^{n-1} = \prod_{j=1}^n \text{Vol}(\pi_j(\bar{S})).$$

Since E is open, $\pi_j(E)$ is open and hence $\pi_j(\overline{S})$ is a subset of $\pi_j(E)$. Thus,

$$(\text{Vol}(\overline{S}))^{n-1} \leq \prod_{j=1}^n \text{Vol}(\pi_j(E))$$

and letting $\epsilon \downarrow 0$, we obtain the result. $\square$

Now we look at the general version of the Loomis-Whitney inequality.

Theorem 2.5 (Loomis-Whitney inequality, general ver.). *Suppose $f_1, \dots, f_n$ are nonnegative functions in $L^1(\mathbb{R}^{n-1})$. Then*

$$\int_{\mathbb{R}^n} \left(\prod_{j=1}^n f_j \circ \pi_j \right)^{\frac{1}{n-1}} \leq \prod_{j=1}^n \left(\int_{\mathbb{R}^{n-1}} f_j \right)^{\frac{1}{n-1}} = \prod_{j=1}^n \|f_j\|_{L^1(\mathbb{R}^{n-1})}^{\frac{1}{n-1}}.$$

Alternatively, if $g_j = f_j^{\frac{1}{n-1}}$ for all $j = 1, \dots, n$, then $g_j \in L^{n-1}(\mathbb{R}^{n-1})$ and

$$\left\| \prod_{j=1}^n g_j \circ \pi_j \right\|_{L^1(\mathbb{R}^n)} \leq \prod_{j=1}^n \|g_j\|_{L^{n-1}(\mathbb{R}^{n-1})}.$$

There are various methods of proving this theorem, such as using the heat flow method (refer to [1]) or the optimal transport method. Here we look at an elementary method that applies induction and uses Hölder's inequality and Fubini's theorem.

Proof. Use induction on n . The base case $n = 2$ follows immediately from Fubini's theorem. Now assume the inequality holds for $n - 1$. Then, applying Theorem 2.1,

$$\begin{aligned} \int_{\mathbb{R}^n} \left(\prod_{j=1}^n f_j \circ \pi_j \right)^{\frac{1}{n-1}} &= \int_{\mathbb{R}^{n-1}} \left((f_n \circ \pi_n)^{\frac{1}{n-1}} \int_{\mathbb{R}} \left(\prod_{j=1}^{n-1} f_j \circ \pi_j \right)^{\frac{1}{n-1}} \right) \\ &\leq \left(\int_{\mathbb{R}^{n-1}} f_n \circ \pi_n \right)^{\frac{1}{n-1}} \left(\int_{\mathbb{R}^{n-1}} \left(\int_{\mathbb{R}} \left(\prod_{j=1}^{n-1} f_j \circ \pi_j \right)^{\frac{1}{n-1}} \right)^{\frac{n-1}{n-2}} \right)^{\frac{n-1}{n-1}} \\ &= \|f_n\|_{L^1(\mathbb{R}^{n-1})}^{\frac{1}{n-1}} \left(\int_{\mathbb{R}^{n-1}} \left(\int_{\mathbb{R}} \left(\prod_{j=1}^{n-1} f_j \circ \pi_j \right)^{\frac{1}{n-1}} \right)^{\frac{n-1}{n-2}} \right)^{\frac{n-2}{n-1}}. \end{aligned}$$

Now applying Theorem 2.2 to the integrand of the second term, we get

$$\left(\int_{\mathbb{R}} \left(\prod_{j=1}^{n-1} f_j \circ \pi_j \right)^{\frac{1}{n-1}} \right)^{\frac{n-1}{n-2}} \leq \left(\prod_{j=1}^{n-1} \int_{\mathbb{R}} f_j \circ \pi_j \right)^{\frac{1}{n-2}}.$$

Hence, by the induction hypothesis,

$$\begin{aligned}
\int_{\mathbb{R}^n} \left(\prod_{j=1}^n f_j \circ \pi_j \right)^{\frac{1}{n-1}} &\leq \|f_n\|_{L^1(\mathbb{R}^{n-1})}^{\frac{1}{n-1}} \left(\int_{\mathbb{R}^{n-1}} \left(\prod_{j=1}^{n-1} \int_{\mathbb{R}} f_j \circ \pi_j \right)^{\frac{1}{n-2}} \right)^{\frac{n-2}{n-1}} \\
&\leq \|f_n\|_{L^1(\mathbb{R}^{n-1})}^{\frac{1}{n-1}} \prod_{j=1}^{n-1} \left(\int_{\mathbb{R}^{n-2}} \left(\int_{\mathbb{R}} f_j \circ \pi_j \right) \right)^{\frac{1}{n-1}} \\
&= \|f_n\|_{L^1(\mathbb{R}^{n-1})}^{\frac{1}{n-1}} \prod_{j=1}^{n-1} \|f_j\|_{L^1(\mathbb{R}^{n-1})}^{\frac{1}{n-1}},
\end{aligned}$$

which completes the proof. $\square$

To see why Theorem 2.5 is the general version of the Loomis-Whitney inequality, we shall deduce Theorem 2.3 using Theorem 2.5. Let E be an open set in $\mathbb{R}^n$ and define functions $f_j = 1_{\pi_j(E)}$ which are nonnegative and are in $L^1(\mathbb{R}^{n-1})$. Since

$$\prod_{j=1}^n (f_j \circ \pi_j)(x) = 1, \quad \text{for all } x \in E,$$

we have that

$$\text{Vol}(E) = \int_{\mathbb{R}^n} 1_E \leq \int_{\mathbb{R}^n} \left(\prod_{j=1}^n f_j \circ \pi_j \right)^{\frac{1}{n-1}} \leq \prod_{j=1}^n \left(\int_{\mathbb{R}^{n-1}} f_j \right)^{\frac{1}{n-1}} = \left(\prod_{j=1}^n \text{Vol}(\pi_j(E)) \right)^{\frac{1}{n-1}}.$$

Hence, Theorem 2.5 is the general case of Theorem 2.3.

We may also discuss whether we can have an exponent other than $\frac{1}{n-1}$ for Theorem 2.5. For instance, suppose the inequality

$$\int_{\mathbb{R}^n} \left(\prod_{j=1}^n f_j \circ \pi_j \right)^p \leq \prod_{j=1}^n \left(\int_{\mathbb{R}^{n-1}} f_j \right)^p \quad (7)$$

holds for all $f_1, \dots, f_n \in L^1(\mathbb{R}^{n-1})$ and $p > 0$. Consider some function $f_j : \mathbb{R}^{n-1} \rightarrow \mathbb{R}$ and define

$$f_j^R(x) = R^{-(n-1)} f_j \left(\frac{x}{R} \right) \quad \text{for all } x \in \mathbb{R}^{n-1} \text{ and } R > 0.$$

Then,

$$\int_{\mathbb{R}^{n-1}} f_j^R(x) dx = \int_{\mathbb{R}^{n-1}} R^{-(n-1)} f_j \left(\frac{x}{R} \right) dx = \int_{\mathbb{R}^{n-1}} f_j \left(\frac{x}{R} \right) d \left(\frac{x}{R} \right) = \int_{\mathbb{R}^{n-1}} f_j(x) dx$$

and

$$\begin{aligned}
\int_{\mathbb{R}^n} \left(\prod_{j=1}^n (f_j^R \circ \pi_j)(x) \right)^p dx &= \int_{\mathbb{R}^n} R^{n-n(n-1)p} \left(\prod_{j=1}^n (f_j \circ \pi_j) \left(\frac{x}{R} \right) \right)^p d \left(\frac{x}{R} \right) \\
&= R^{n-n(n-1)p} \int_{\mathbb{R}^n} \left(\prod_{j=1}^n (f_j \circ \pi_j)(x) \right)^p dx.
\end{aligned}$$

Since $f_j \in L^1(\mathbb{R}^{n-1})$ implies $f_j^R \in L^1(\mathbb{R}^{n-1})$, (7) would imply the inequality

$$R^{n-n(n-1)p} \int_{\mathbb{R}^n} \left(\prod_{j=1}^n f_j \circ \pi_j \right)^p \leq \prod_{j=1}^n \left(\int_{\mathbb{R}^{n-1}} f_j \right)^p,$$

which is a contradiction for $p < \frac{1}{n-1}$ when $R > 1$ and $p > \frac{1}{n-1}$ when $R < 1$. Thus, inequalities of the form (7) hold only for $p = \frac{1}{n-1}$, which is our Loomis-Whitney inequality.

2.2 Relation to Theorem 1.5

The Loomis-Whitney inequality is a special case of Theorem 1.5 and (5) when the given cylinders are all parallel to some axis of $\mathbb{R}^n$ [1, 7]. Here we adapt the notation $a \lesssim_n b$ for $a, b \in \mathbb{R}$ to mean that there exists some constant $C_n \geq 0$ depending on n in $\mathbb{R}$ such that $a \leq C_n b$.

Return to the assumptions in Theorem 1.5: let $T_{j,a}$ be cylinders of radius 1 in $\mathbb{R}^n$, where $1 \leq j \leq n$ and $1 \leq a \leq A$. If $T_{j,a}$ are exactly parallel to the x_j -axis, then

$$\text{Vol}(\pi_j(I)) = \text{Vol} \left(\pi_j \left(\bigcap_{j=1}^n \bigcup_{a=1}^A T_{j,a} \right) \right) \lesssim_n A(j)$$

since $\pi_j(I)$ is contained in $A(j)$ unit balls. By applying Theorem 2.3, we have

$$\text{Vol}(I) \leq \prod_{j=1}^n \text{Vol}(\pi_j(I))^{\frac{1}{n-1}} \lesssim_n \prod_{j=1}^n A^{\frac{1}{n-1}} = A^{\frac{n}{n-1}},$$

the same result as in Theorem 1.5.

If in general $1 \leq a \leq A(j)$, then similarly,

$$\text{Vol}(I) \lesssim_n \prod_{j=1}^n A(j)^{\frac{1}{n-1}},$$

which is consistent with (3) due to (6). This proves that the endpoint case of the multilinear Kakeya estimate is a generalisation of the Loomis-Whitney inequality.

3 Borsuk-Ulam theorem

The main tool we will need to prove Theorem 1.5 is the so-called polynomial ham sandwich theorem, which follows almost immediately from a more well-known result from topology called the Borsuk-Ulam theorem. In this section, we give a proof of the Borsuk-Ulam theorem in N dimensions. The exposition here draws from [10].

The Borsuk-Ulam theorem is a statement about continuous maps from spheres to hyperplanes. Namely, it states that if $F : S^N \rightarrow \mathbb{R}^N$ is a continuous map, it must have two antipodal points on the sphere identified by the map. That is, there exists a ξ such that $F(\xi) = F(-\xi)$. The way we will prove this is by considering the differences between where antipodal points land. If no such point ξ exists, then the difference $F(x) - F(-x)$ would be non-zero for every point x on the sphere. In other words, there is a positive distance between the points $F(x)$

and $F(-x)$. We shall see that if this is true, it would induce a very special continuous map of spheres, $g : S^N \rightarrow S^{N-1}$, for which antipodal points on S^N have antipodal images on S^{N-1} .

A map which keeps antipodal points antipodal is equivalent to a map of real projective spaces, spaces formed by glueing together antipodal points and considering them a single point. The key to proving the Borsuk-Ulam theorem will be to establish a sphere S^N as a covering space of real projective space $\mathbb{R}P^N$, and lift a mapping of real projective spaces to a mapping onto the sphere, its covering space. Thus, we begin by discussing covering spaces.

3.1 Covering spaces

In this subsection, we introduce some general definitions and theory of covering spaces, and in the next subsection, we move on to the specific case of spheres and real projective spaces, which are needed to prove the Borsuk-Ulam theorem.

Definition 3.1 (Fibre). Let X, Y be topological spaces, and $f : X \rightarrow Y$ be a continuous map. For any $y \in Y$, we call $f^{-1}(\{y\})$ the *fibre over y* , denoted by $f^{-1}(y)$.

Definition 3.2 (Lifting). Let X, Y, Z be topological spaces. Let $f : X \rightarrow Z$, $g : Y \rightarrow Z$, $h : X \rightarrow Y$ be continuous maps. We say h is a *lifting* of f to Y if $f = g \circ h$.

$$\begin{array}{ccc} & & Y \\ & \nearrow h & \downarrow g \\ X & \xrightarrow{f} & Z \end{array}$$

Definition 3.3 (Even covering). Let $X, \tilde{X}$ be topological spaces, and $p : \tilde{X} \rightarrow X$ a continuous map. We say that an open set U of X is *evenly covered* by p if there exist disjoint open sets S_i of $\tilde{X}$ such that $p^{-1}(U) = \bigsqcup S_i$, and for each i the restriction $p|_{S_i} : S_i \rightarrow U$ is a homeomorphism.

In other words, the preimage of the open set U under the (potentially non-injective) continuous map p is partitioned into *sheets* S_i , each of which maps perfectly onto U under p .

Example 3.4. Let S^2 denote the unit sphere in $\mathbb{R}^3$, and let D^2 denote the open unit disc in $\mathbb{R}^2$. D^2 is evenly covered by $p : S^2 \rightarrow D^2$ given by $p(x, y, z) = (x, y)$.

Proof. We can let S_1 and S_2 be the top and bottom hemispheres of the unit sphere, respectively. It is then easy to see that the restriction of p to each hemisphere is a homeomorphism onto the unit disc, and the preimage of the unit disc is precisely the sphere minus its equator, that is, the union of S_1 and S_2 . $\square$

Definition 3.5 (Covering space). Let $X, \tilde{X}$ be path-connected topological spaces, and $p : \tilde{X} \rightarrow X$ a continuous map. We call $(\tilde{X}, p)$ a *covering space* of X if every point $x \in X$ has a neighbourhood U_x that is evenly covered by p .

The map p in this context is called a *covering projection*.

Some texts would call this a *path-connected covering*. We will, however, only consider path-connected spaces and so will not make any distinction.

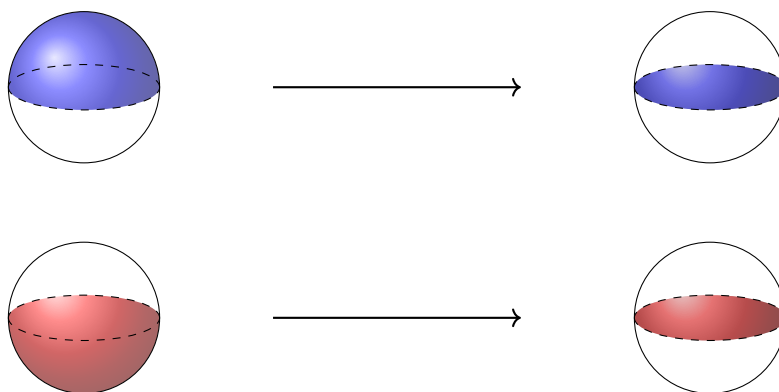


Figure 1 Example 3.4: The homeomorphisms $S_1 \rightarrow D^2$ and $S_2 \rightarrow D^2$ obtained by restricting p .

A covering space globalises the idea of an even covering. A particular subset may be evenly covered, but a covering space says that a topological space is locally evenly covered in the same way at every point.

The following example is the classic one.

Example 3.6. Let S^1 denote the unit circle in $\mathbb{C}$. Let $p : \mathbb{R} \rightarrow S^1$ be the map $t \mapsto \exp(2\pi it)$. Then, $(\mathbb{R}, p)$ is a covering space of S^1 .

Proof. We can think of p winding each unit interval $[i, i + 1)$ around the circle, so that $\mathbb{R}$ is essentially infinitely many unwinded copies of S^1 .

Let $\theta \in [0, 1)$ be arbitrary. Then, $x = \exp(2\pi i\theta)$ is an arbitrary point on S^1 . $p((\theta - 0.2, \theta + 0.2))$ is a neighbourhood of S^1 containing x , and it is evenly covered by p . $\square$

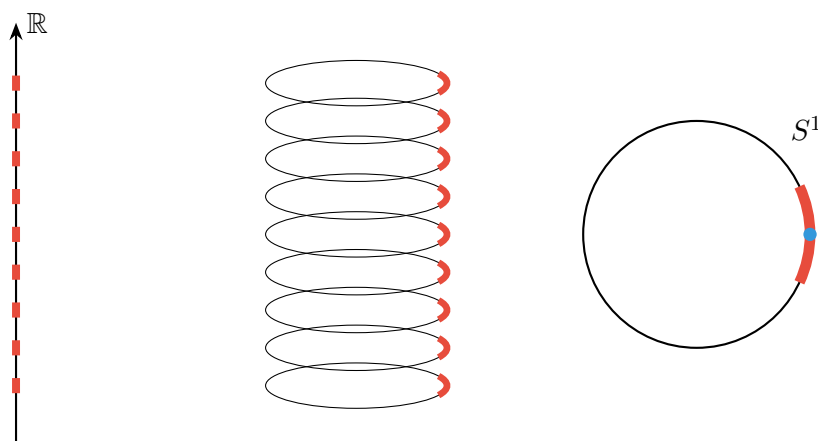


Figure 2 Example 3.6: A diagram illustrating the covering projection $p : \mathbb{R} \rightarrow S^1$.

Like in the above example, it is often helpful to think of a covering space as multiple copies of the space it covers.

Definition 3.7 (Pointed space). A *pointed space* (X, x_0) is a topological space X along with a point $x_0 \in X$, called the *basepoint*.

If $(X, x_0), (Y, y_0)$ are pointed spaces, $f : (X, x_0) \rightarrow (Y, y_0)$ is a map between pointed spaces if $f : X \rightarrow Y$ is a mapping of topological spaces and $f(x_0) = y_0$.

The definitions we have made thus far have pointed space analogues, where the mappings must preserve the basepoint.

Lemma 3.8 (Uniqueness of lifting to a covering space). *Let $(X, x_0), (Y, y_0)$ be pointed spaces, $f : (Y, y_0) \rightarrow (X, x_0)$ a continuous map, and $(\tilde{X}, p)$ a covering space of X . If Y is connected, then there exists at most one lifting of f to $(\tilde{X}, \tilde{x}_0)$.*

Proof. We argue by contraposition. Assume there are two liftings $\tilde{f}_1 : (Y, y_0) \rightarrow (\tilde{X}, \tilde{x}_0)$ and $\tilde{f}_2 : (Y, y_0) \rightarrow (\tilde{X}, \tilde{x}_0)$ such that $\tilde{f}_1 \neq \tilde{f}_2$. We aim to find two open subsets $A, B \subseteq Y$ such that $Y = A \cup B$, $A \cap B = \emptyset$, $A \neq \emptyset$ and $B \neq \emptyset$.

Let $A = \{a \in Y \mid \tilde{f}_1(a) \neq \tilde{f}_2(a)\}$ and $B = Y \setminus A$. We know that $A \neq \emptyset$ since $\tilde{f}_1 \neq \tilde{f}_2$, and we know that $B \neq \emptyset$ since both are maps from (Y, y_0) to $(\tilde{X}, \tilde{x}_0)$, so $y_0 \in B$. Clearly $A \cup B = Y$ and $A \cap B = \emptyset$. Thus, it remains only to show that A and B are open in Y .

Let $a \in A$ be arbitrary. $(\tilde{X}, p)$ is a covering space of X so there exists a neighbourhood U_a of X containing $f(a)$ such that U_a is evenly covered by p . That is, there exist sheets S_i such that $p^{-1}(U_a) = \bigsqcup S_i$ and $p|_{S_i} : S_i \rightarrow U_a$ are homeomorphisms. We know $p\tilde{f}_1(a) = p\tilde{f}_2(a) = f(a)$. Since the restriction of p to any sheet is injective, if $\tilde{f}_1(a)$ and $\tilde{f}_2(a)$ were in the same sheet, it would force $\tilde{f}_1(a) = \tilde{f}_2(a)$, but since $a \in A$ we know that this is not the case. Therefore they must be in distinct sheets, say $\tilde{f}_1(a) \in S_i$ and $\tilde{f}_2(a) \in S_j$, where $i \neq j$. Consider the preimages of these sheets, $\tilde{f}_1^{-1}(S_i)$ and $\tilde{f}_2^{-1}(S_j)$. a is in both of these, i.e. $a \in \tilde{f}_1^{-1}(S_i) \cap \tilde{f}_2^{-1}(S_j)$. In fact, since S_i and S_j are disjoint, any member of this intersection is in A . Also, this intersection is an open set since the sheets are open, $\tilde{f}_1$ and $\tilde{f}_2$ are continuous, and the intersection of two open sets is open. Therefore, $\tilde{f}_1^{-1}(S_i) \cap \tilde{f}_2^{-1}(S_j)$ is an open neighbourhood in A containing a , meaning A is open.

Let $b \in B$ be arbitrary. Let U_b be a neighbourhood of X containing $f(b)$, and let the preimage be $p^{-1}(U_b) = \bigsqcup S_i$. Let S be the sheet containing $\tilde{f}_1(b)$ and $\tilde{f}_2(b)$. Note that this time they are in the same sheet because they are equal. We consider the intersection $\tilde{f}_1^{-1}(S) \cap \tilde{f}_2^{-1}(S)$. We know that b is in this intersection. Also, any member y of the intersection satisfies $\tilde{f}_1(y), \tilde{f}_2(y) \in S$, and since $p|_S$ is injective and $p\tilde{f}_1(y) = p\tilde{f}_2(y) = f(y)$, it must be that $\tilde{f}_1(y) = \tilde{f}_2(y)$. Also, as before, the intersection is an open set. Therefore, $\tilde{f}_1^{-1}(S) \cap \tilde{f}_2^{-1}(S)$ is an open neighbourhood in B containing b , meaning B is open. $\square$

3.2 Proof of Borsuk-Ulam

Now that we have covered all of the general definitions and theory of covering spaces we need, we can begin talking about the specific spaces that are relevant to proving the Borsuk-Ulam theorem.

Definition 3.9 (Sphere). Denote by S^N the unit sphere in $\mathbb{R}^{N+1}$. That is,

$$S^N = \{(x_0, \dots, x_N) \in \mathbb{R}^{N+1} \mid x_0^2 + \dots + x_N^2 = 1\}.$$

Definition 3.10 (Antipodal). For any N , we say that points $x, y \in S^N$ are *antipodal* if $y = -x$.

Definition 3.11 (Real projective space). We define *real projective space of dimension N* to be S^N relating antipodal points. That is, we define an equivalence relation on S^N

$$x \sim y \iff (x = y \text{ or } x = -y),$$

then we define

$$\mathbb{R}P^N = S^N / \sim.$$

$\mathbb{R}P^N$ is endowed with the quotient topology, meaning a set $U \subseteq \mathbb{R}P^N$ is open if and only if $v^{-1}(U)$ is open in S^N , where $v : S^N \rightarrow \mathbb{R}P^N$ is the canonical map $x \mapsto [x]$.

Proposition 3.12 (Covering projection identifying antipodal points). *Let $v : S^N \rightarrow \mathbb{R}P^N$ be the canonical map $x \mapsto [x]$. Then (S^N, v) is a covering space of $\mathbb{R}P^N$, and for any distinct points $x, y \in S^N$, $v(x) = v(y)$ if and only if x and y are antipodal.*

Proof. Let $x, y \in S^N$ be distinct points. $v(x) = \{x, -x\}$ and $v(y) = \{y, -y\}$. So $v(x) = v(y)$ if and only if $-x = y$, i.e. x and y are antipodal.

If S is a set, we use the notation $-S$ to denote the set $\{-s : s \in S\}$.

Let $[x] \in \mathbb{R}P^N$ be arbitrary. Let V_x be the set of points on S^N for which the angle between the line segment joining 0 and the point and the line segment joining 0 and x is strictly less than 0.1. V_x is a neighbourhood of S^N containing x . Then let $S_1 = V_x$ and $S_2 = -V_x$. These are disjoint. The images of S_1 and S_2 are the same under v . Call this common image U_x . The preimage $v^{-1}(U_x)$ is equal to the union of S_1 and S_2 , which are both open. So by the definition of the topology on $\mathbb{R}P^N$, U_x is open in $\mathbb{R}P^N$.

The restrictions $v|_{S_1}$ and $v|_{S_2}$ are bijective onto U_x , and by the definition of the topology on $\mathbb{R}P^N$, they are continuous. We need to show that their inverses are also continuous.

Let $i \in \{1, 2\}$ be arbitrary. Let V be an open subset of S_i . We know its preimage $(v|_{S_i}^{-1})(V) = V \cup -V$ is the union of two open sets, which is open. Thus $v|_{S_i}^{-1}$ is continuous. This proves that the restrictions are homeomorphisms $S_i \rightarrow U_x$, and thus (S^N, v) is a covering space of $\mathbb{R}P^N$. $\square$

Lemma 3.13. *Let $n > m \geq 1$, and let $f : \mathbb{R}P^n \rightarrow \mathbb{R}P^m$ be a continuous map. If $v : S^m \rightarrow \mathbb{R}P^m$ is the covering projection identifying antipodal points, then there exists a lifting $\tilde{f} : \mathbb{R}P^n \rightarrow S^m$ (i.e., $v \circ \tilde{f} = f$).*

$$\begin{array}{ccc} & & S^m \\ & \nearrow \tilde{f} & \downarrow v \\ \mathbb{R}P^n & \xrightarrow{f} & \mathbb{R}P^m \end{array}$$

Proof. The proof of this lemma requires more sophisticated algebraic topology and is too long to be covered here. The interested reader may be directed to chapter 12 of [10] for a full discussion. $\square$

Theorem 3.14. *Let $n > m \geq 1$, and let $g : S^n \rightarrow S^m$ be continuous. Then there exists at least one point $y \in S^n$ such that $g(-y) \neq -g(y)$.*

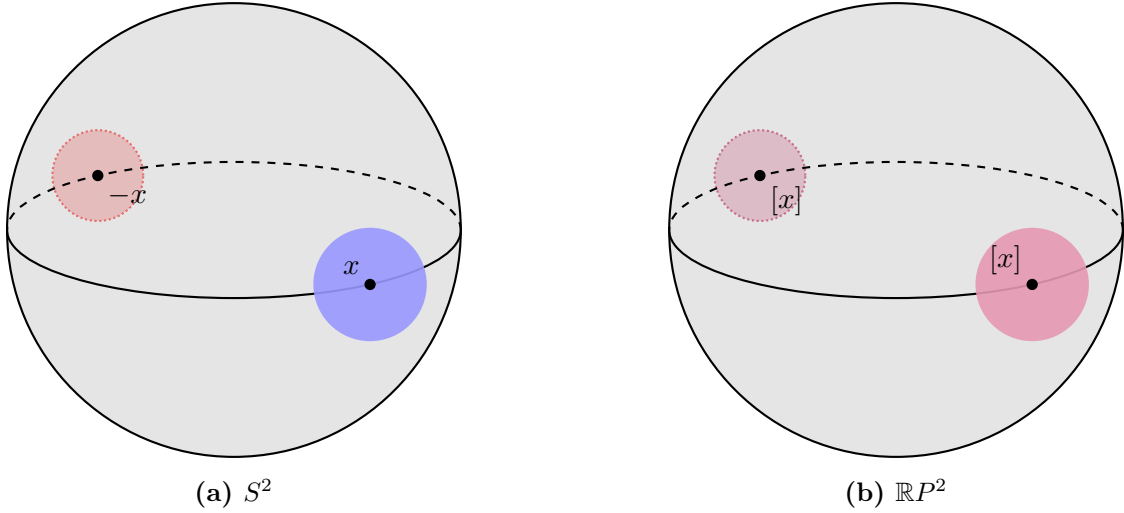


Figure 3 Proposition 3.12: The covering projection $v : S^2 \rightarrow \mathbb{R}P^2$. On the right is the neighbourhood of $[x]$. On the left is the preimage $S_1 \sqcup S_2$, with S_1 in blue at the front and S_2 in red at the back.

Proof. Suppose, for contradiction, that for all $y \in S^n$, $g(-y) = -g(y)$. This induces the continuous map $f : \mathbb{R}P^n \rightarrow \mathbb{R}P^m$ which sends a point $[y] \in \mathbb{R}P^n$ to $[g(y/\|y\|)]$. This is well defined since if $[y] = [x]$ then either $x/\|x\| = y/\|y\|$, in which case $f([x]) = f([y])$, or, $x/\|x\| = -y/\|y\|$, in which case $f([x]) = [g(-y/\|y\|)] = [-g(y/\|y\|)] = [g(y/\|y\|)] = f([y])$.

Let $v' : S^n \rightarrow \mathbb{R}P^n$ and $v : S^m \rightarrow \mathbb{R}P^m$ be the covering projections identifying antipodal points. Then, the induced map f defined above satisfies $f \circ v' = v \circ g$. In other words, f makes the following diagram commute:

$$\begin{array}{ccc} S^n & \xrightarrow{g} & S^m \\ v' \downarrow & & \downarrow v \\ \mathbb{R}P^n & \xrightarrow{f} & \mathbb{R}P^m \end{array} .$$

By Lemma 3.13, there exists a lifting of f to S^m :

$$\begin{array}{ccc} & & S^m \\ & \nearrow \tilde{f} & \downarrow v \\ \mathbb{R}P^n & \xrightarrow{f} & \mathbb{R}P^m \end{array} ,$$

and right composition with v' yields $f \circ v' = v \circ \tilde{f} \circ v'$.

Overall, we see that both g and $\tilde{f} \circ v'$ are liftings of $f \circ v'$ to S^m :

$$\begin{array}{ccc} & & S^m \\ & \nearrow g & \downarrow v \\ S^n & \xrightarrow{f \circ v'} & \mathbb{R}P^m \end{array} \qquad \begin{array}{ccc} & & S^m \\ & \nearrow \tilde{f} \circ v' & \downarrow v \\ S^n & \xrightarrow{f \circ v'} & \mathbb{R}P^m \end{array} .$$

We will now apply the uniqueness lemma (Lemma 3.8) to argue that $g = \tilde{f} \circ v'$: choose some $y_0 \in S^n$ and let $x_0 = (f \circ v')(y_0)$. We know v is a two-to-one mapping from S^m to $\mathbb{R}P^m$ that identifies antipodal points, and the diagram on the left shows that $g(y_0)$ is mapped to x_0 ,

which means that the only other point that maps to x_0 is $-g(y_0)$. In other words, the fibre over x_0 is $v^{-1}(x_0) = \{g(y_0), -g(y_0)\}$. By the property of g , this means $v^{-1}(x_0) = \{g(y_0), g(-y_0)\}$. Therefore, $(\tilde{f} \circ v')(y_0) = g(\pm y_0)$. Suppose that $(\tilde{f} \circ v')(y_0) = g(-y_0)$. Then, since v' identifies antipodal points, $(\tilde{f} \circ v')(-y_0) = (\tilde{f} \circ v')(y_0) = g(-y_0)$. So either $(\tilde{f} \circ v')(y_0) = g(y_0)$ or $(\tilde{f} \circ v')(-y_0) = g(-y_0)$. Without loss of generality, say $(\tilde{f} \circ v')(y_0) = g(y_0)$. Then $f \circ v'$ is a continuous map of pointed spaces $(S^n, y_0) \rightarrow (\mathbb{R}P^m, x_0)$ and both g and $\tilde{f} \circ v'$ are liftings to $(S^m, g(y_0))$. Thus, since S^n is connected, we can apply the uniqueness lemma (Lemma 3.8) to conclude $g = \tilde{f} \circ v'$.

This is a problem. v' identifies antipodal points, meaning for all $y \in S^n$, $(\tilde{f} \circ v')(-y) = (\tilde{f} \circ v')(y)$. However, the property of g would imply that for all $y \in S^n$, $(\tilde{f} \circ v')(-y) = -(\tilde{f} \circ v')(y)$. Since no point on S^m satisfies $x = -x$, this is a contradiction. $\square$

Corollary 3.15 (Borsuk-Ulam Theorem). *For any $N \geq 1$, if $F : S^N \rightarrow \mathbb{R}^N$ is continuous, then there exists a point $\xi \in S^N$ such that $F(\xi) = F(-\xi)$.*

Proof. Suppose, for contradiction, that no such point exists. Then we can define a map $g : S^N \rightarrow S^{N-1}$ as

$$g(x) = \frac{F(x) - F(-x)}{\|F(x) - F(-x)\|}.$$

By the assumptions, this is seen to be a continuous map, and it satisfies $g(-x) = -g(x)$ for all $x \in S^N$. This contradicts Theorem 3.14. $\square$

4 Polynomial ham sandwich theorem

4.1 Classical ham sandwich theorem

We first present the classical case of the ham sandwich theorem, which tells us that for n measurable objects in $\mathbb{R}^n$ there exists a hyperplane which bisects every object such that the area above the hyperplane is equal to the area below the hyperplane.

Theorem 4.1 (Classical ham sandwich theorem). *Let U_i , where $1 \leq i \leq n$, be (Lebesgue) measurable sets in $\mathbb{R}^n$. There exists a hyperplane H that bisects every U_i .*

The general proof method of this theorem relies on the Borsuk-Ulam theorem (Theorem 3.15) and also on an initial application of the intermediate value theorem. The idea of this proof generalises to more complex surfaces, which we will soon explore.

Proof. We first consider any point $p \in S^{n-1}$. Where $S^{n-1} = \{x \in \mathbb{R}^n \mid \|x\|_2 = 1\}$. Geometrically, if we interpret the point p as a normal vector, we can associate to every point p in S^{n-1} , a family of hyperplanes $H_p = \{H'_p + tp \mid t \in \mathbb{R}\}$ where $H'_p = \{x \in \mathbb{R}^n \mid x \cdot p = 0\}$.

We now consider U_1 . Fix $p \in S^{n-1}$, intuitively we take one side of our hyperplane as positive, and move the hyperplane through Euclidean space by slowly increasing t , looking at the volume of intersection of U_1 and upper-half of the plane, we stop changing t when we reach half the volume of U_1 . We know we can do this by the mean value theorem.

Formally, define $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(t) = \text{Vol}(\{x \in H_p \mid x \cdot p > 0\} \cap U_1)$. We see that the hyperplane H_p depends continuously on t , thus f is continuous and hence by the intermediate value theorem, there is some ξ such that $f(\xi) = \frac{1}{2} \text{Vol}(U_1)$. So we have shown that for any point $p \in S^{n-1}$ we have a unique hyperplane which bisects U_1 .

We now always refer to this unique hyperplane for p by choosing $t = \xi$ as H_p . Consider the other $n - 1$ objects $U_2, \dots, U_n$. Define a new function $g : S^{n-1} \rightarrow \mathbb{R}^{n-1}$ where $g_i(p) = \text{Vol}(\{x \in H_p \mid x \cdot p > 0\} \cap U_{i+1}) - \text{Vol}(\{x \in H_p \mid x \cdot p < 0\} \cap U_{i+1})$ denotes the i^{th} component of g .

We see that interchanging p with $-p$ merely changes the side of our plane we regard as positive, and the two volumes above trade places. Thus we have that g is an odd function. It is again clear that g is continuous, since it is continuous in each component. Now we apply Borsuk-Ulam theorem (Theorem 3.15) to get that there exists $p \in S^{n-1}$ such that $g(p) = g(-p)$ implies $g(p) = 0$ and thus the hyperplane H_p must bisect $U_2, \dots, U_n$ and also U_1 by previous arguments, which gives the required result. $\square$

4.2 Proof of the polynomial ham sandwich theorem

One may wonder if we can do even better than the above theorem, for example, can we have some hypersurface that could bisect $n + 1$ objects in $\mathbb{R}^n$? The answer is yes, however we must consider now some perturbed hypersurface rather than a mere hyperplane. We introduce the following definitions in preparation for the main theorem.

Definition 4.2 (Polynomial of degree d in n variables). In general, a polynomial of degree d in n variables is defined as:

$$P(x_1, x_2, \dots, x_n) = \sum_{\alpha \in \mathbb{N}^n, |\alpha| \leq d} c_\alpha x^\alpha$$

where $x^\alpha = x_1^{\alpha_1} \dots x_n^{\alpha_n}$ and $|\alpha| = \alpha_1 + \dots + \alpha_n$.

Definition 4.3 (Affine algebraic hypersurface of degree d). An *(affine) algebraic hypersurface* of degree d in $\mathbb{R}^n$ is the following set:

$$Z = \{x \in \mathbb{R}^n \mid f(x_1, \dots, x_n) = 0\}$$

where f is a polynomial of degree d . This is also called a *polynomial surface*.

An important part of proving the polynomial version of the polynomial ham sandwich theorem will be to understand the vector space $V(d)$, the set of all polynomials $P : \mathbb{R}^n \rightarrow \mathbb{R}$ with degree no greater than d . The lemma below gives a simple combinatorial justification for the dimension of this vector space.

Lemma 4.4. $\dim(V(d)) = \binom{n+d}{d}$

Proof. One notes that any polynomial in the vector space is specified by a linear sum of its monomial terms x^α . It is clear that distinct monomial terms are linearly independent and span our vector space. So it remains to determine how many such monomial terms exist. Thus the problem reduces to finding number of tuples $\alpha = (a_1, \dots, a_n) \in \mathbb{N}^n \cup \{0\}$ such that $|\alpha| = \alpha_1 + \dots + \alpha_n \leq d$. Upon introducing a dummy variable $\alpha_{n+1} \in \mathbb{N} \cup \{0\}$ The problem is

to determine number of solutions to $\alpha_1 + \dots + \alpha_{n+1} = d$ which is given by a simple stars and bars argument as $\binom{n+d}{d}$. $\square$

Finally we are ready to state and prove the polynomial ham-sandwich theorem. The ideas of the following proof are an explained and adapted version of the proof given by Larry Guth in [6].

Theorem 4.5 (Polynomial Ham Sandwich Theorem). *Let $N = \binom{n+d}{d} - 1$. Let $U_1, \dots, U_N$ be finite volume open sets in $\mathbb{R}^n$. Then there exists a real algebraic hypersurface Z of degree at most d which bisects each set U_i .*

Proof. We aim to set up a continuous function so that we can apply the Borsuk-Ulam theorem. The idea is nearly identical to the case above.

First, we note that every $P \in S^N$ specifies a unique algebraic hypersurface of degree at most d , namely, the algebraic hypersurface defined by a polynomial with coefficients given by P . We refer to this polynomial as P' . Next we construct the function $f : S^N \rightarrow \mathbb{R}^N$ where $f_i(P) = \text{Vol}\{P'(x) > 0 \cap U_i\} - \text{Vol}\{P'(x) < 0 \cap U_i\}$ denotes the i^{th} component of f . Again, note that f is odd since $f(-P)$ gives the same algebraic hypersurface, yet now the volumes in the definition of f_i switch places. We can see that each f_i is continuous since a small change in measure can be easily achieved by a small change in our coefficient variables for P (we will formalise this below). Thus the result follows from the Borsuk-Ulam theorem (Theorem 3.15), by the same considerations as the classical version above. $\square$

We note that by setting some of the sets U_i (as defined above) to be empty, we can consider any number of sets U_i in $\mathbb{R}^n$, and as long as we choose a large enough degree d , we can construct the required algebraic hypersurface.

Finally, we give a formal proof that f_i is continuous to finish the theorem.

Lemma 4.6. *Let U be an open (Lebesgue) measurable set in $\mathbb{R}^n$. The function $f_i : S^N \rightarrow \mathbb{R}$ where $f_i(P) = \text{Vol}\{P'(x) > 0 \cap U\} - \text{Vol}\{P'(x) < 0 \cap U\}$ is continuous on all points in domain.*

Proof. We show this via a sequential continuity argument. This is equivalent to the topological continuity in Borsuk-Ulam since both S^N and $\mathbb{R}^N$ are standard topologies inherited from the usual metrics on these spaces.

We assume $P_n \rightarrow P$ in S^N . Under Euclidean metric we see this means that the coefficients of $P'_n(x)$ become arbitrarily close to $P'(x)$ for large n and thus $P'_n(x) \rightarrow P'(x)$ pointwise on $\mathbb{R}^n \supset U$. Now we apply Egorov's theorem to get $E \subset U$ where $\text{Vol}(E) = \epsilon$ so that $P'_n(x) \rightarrow P'(x)$ uniformly on $U - E$.

Since $\text{Vol}\{P'(x) = 0\} = 0$ intuitively we can choose δ small enough such that $\text{Vol}\{|P'(x)| < \delta\} = \epsilon$. Since $P'_n(x) \rightarrow P'(x)$ uniformly on $U - E$, we can choose n large enough to bring the polynomial surface P'_n within δ of $P'(x)$. And so we note that for large enough n the measures $\text{Vol}\{P'(x) > 0 \cap U\}$ and $\text{Vol}\{P'_n(x) > 0 \cap U\}$ differ by no more than 2ϵ which finishes our sequential continuity proof and gives required result. $\square$

5 Cylinder estimate

Definition 5.1. A cylinder $T \in \mathbb{R}^n$ with ‘core’ v and ‘radius’ r (through the origin) is defined as:

$$T = \{x \in \mathbb{R}^n \mid \text{dist}(x, L_v) \leq r\}$$

where $\text{dist} : \mathbb{R}^n \rightarrow \mathbb{R}$ is defined by $\text{dist}(x, S) = \inf\{d(x, v) \mid v \in S\}$ for $x \in \mathbb{R}^n$ and $S \subset \mathbb{R}^n$; L_v is the line $\text{span}(v)$.

This coincides with our intuition in $\mathbb{R}^3$ where we note that the ‘core’ of the cylinder is simply just the line that runs through the centre of the cylinder. The above definition extends this idea to any arbitrary dimension. For an arbitrary cylinder not through the origin, we simply translate this set by any constant vector $c \in \mathbb{R}^n$.

Definition 5.2 (Directed volume). For a hypersurface $S \subset \mathbb{R}^n$, the directed volume function V_S is given by the following:

$$V_S(v) = \int_S |v \cdot N| \, d\text{vol}_S$$

Where N denote the normal vector to S , and $v \in \mathbb{R}^n$

For unit vector v , let $\pi_v : \mathbb{R}^n \rightarrow v^\perp$, the directed volume can also be considered as the volume of $\pi_v(S)$, counting geometric multiplicity. So it can be seen as:

$$V_S(v) = \int_{v^\perp} |S \cap \pi_v^{-1}(y)| \, dy$$

To show that these definitions are equivalent for unit vectors, we consider the projection of $\int_S |v \cdot N| \, d\text{vol}_S$ onto $v^\perp$. By projection theorem, we have for each point $x \in S$, we have $|v \cdot N(x)| \, d\text{vol}_S = d\text{vol}_{\pi_v(S)}$, but it is possible that multiple points in S projecting onto the same point $y \in v^\perp$, so we have to multiply by $|S \cap \pi_v^{-1}(y)|$ to count the geometric multiplicity. Thus:

$$\int_S |v \cdot N| \, d\text{vol}_S = \int_{\pi_v(S)} |S \cap \pi_v^{-1}(y)| \, dy$$

and the function is supported only within $\pi_v(S)$, so we have shown that they are indeed equivalent

Lemma 5.3 (Cylinder estimate). *If T is a cylinder of radius r , v is a unit vector parallel to the core of T , and Z is an algebraic hypersurface of degree d , then the directed volume $V_{Z \cap T}(v)$ is bounded as follows:*

$$V_{Z \cap T}(v) \leq \omega_{n-1} r^{n-1} d$$

where ω_n is the volume of n -dimensional unit ball.

Proof. To evaluate $V_{Z \cap T}(v)$, first we have that $\pi_v(T)$ is a $n-1$ dimensional ball of radius r , and we must have $|Z \cap T \cap \pi_v^{-1}(y)| = 0$ for $y \notin \pi_v(T)$, where $y \in v^\perp$, thus we have:

$$\begin{aligned} V_{Z \cap T}(v) &= \int_{v^\perp} |Z \cap T \cap \pi_v^{-1}(y)| \, dy \\ &= \int_{\pi_v(T)} |Z \cap T \cap \pi_v^{-1}(y)| \, dy \\ &= \int_{\pi_v(T)} |Z \cap \pi_v^{-1}(y)| \, dy \end{aligned}$$

Notice that $Z \cap \pi_v^{-1}(y)$ are the roots of polynomial of degree d , thus either $|Z \cap \pi_v^{-1}(y)| \leq d$ or $|Z \cap \pi_v^{-1}(y)| = \infty$, but the second case can only occur with 0 measure. Combining with $\int_{\pi_v(T)} dy = \omega_{n-1} r^{n-1}$, the result is obvious. $\square$

Lemma 5.4 (Summation of directed volumes). *Suppose that $v_1, \dots, v_n$ are unit vectors. Let e_j denote the coordinate unit vectors and suppose that $\|e_j - v_j\| < (100n)^{-1}$. Let S be any hypersurface in $\mathbb{R}^n$. Then*

$$\text{Vol}(S) \leq 2 \sum_{j=1}^n V_S(v_j).$$

Proof. Let $N(x)$ denote the unit normal vector to S at x . We start by considering $V_S(v_j) = \int_S |v_j \cdot N(x)| d\text{vol}_S$. By Cauchy-Schwarz:

$$|v_j \cdot N(x) - e_j \cdot N(x)| = |(v_j - e_j) \cdot N(x)| \leq \|v_j - e_j\| \|N(x)\| < (100n)^{-1}.$$

Thus, $|v_j \cdot N(x)| > |e_j \cdot N(x)| - (100n)^{-1}$. Summing over j gives

$$\sum_{j=1}^n |v_j \cdot N(x)| > \sum_{j=1}^n |e_j \cdot N(x)| - \frac{1}{100} \geq \frac{99}{100}.$$

The last inequality holds because $\sum_{j=1}^n |e_j \cdot N(x)| = \sum_{j=1}^n |N(x)_j| \geq \sum_{j=1}^n N(x)_j^2 = 1$, where $N(x)_j$ denotes the j^{th} component of $N(x)$, so we have

$$\sum_{j=1}^n V_S(v_j) = \sum_{j=1}^n \int_S |v_j \cdot N(x)| d\text{vol}_S \geq \int_S \frac{99}{100} d\text{vol}_S = \frac{99}{100} \text{Vol}(S).$$

The result follows by rearranging and taking less strict bound. $\square$

6 Proof of Theorem 1.5

Recall Theorem 1.5, the weak endpoint case of the multilinear Kakeya estimate:

Theorem 6.1. *Let $T_{j,a}$ be arbitrary cylinders in $\mathbb{R}^n$ with radius 1 and infinite length, ranging over $1 \leq j \leq n$ and $1 \leq a \leq A$. If $T_{j,a}$ are parallel to some direction within an angle of $(100n)^{-1}$ to the x_j -axis and*

$$I = \bigcap_{j=1}^n \bigcup_{a=1}^A T_{j,a},$$

then there exists some constant C_n such that $\text{Vol}(I) \leq C_n A^{\frac{n}{n-1}}$.

Our final important lemma gives us an upper bound on our choice of degree d from the polynomial ham sandwich theorem. We want to know, given V objects in $\mathbb{R}^n$, how large we have to make the degree d to find an algebraic hypersurface that bisects all V objects.

Lemma 6.2. *Given V objects in $\mathbb{R}^n$, there exists an algebraic hypersurface of degree d , where $d \lesssim V^{1/n}$, that bisects these objects.*

Proof. We increase d until $V < d^n/n! \leq (d+1) \cdots (d+n)/n! = \binom{n+d}{d}$. Then, by the polynomial ham sandwich theorem (Theorem 4.5), we know that any d satisfying the above inequality will work. Rearranging gives any d , where $d > (n!V)^{1/n}$, will work. Therefore, choosing $d = \lceil (n!V)^{1/n} \rceil \leq (n!V)^{1/n} + 1 \leq 2(n!)^{1/n}V^{1/n} \lesssim V^{1/n}$ works as required. $\square$

We present a justification of the basic area estimate, which will also be used in our proof of Theorem 1.5.

Definition 6.3. (Minkowski addition) In geometry, the *Minkowski sum* of two sets of position vectors A and B in $\mathbb{R}^n$ is formed by adding each vector in A to each vector in B :

$$A + B = \{a + b \mid a \in A, b \in B\}.$$

Definition 6.4. (d -dimensional Hausdorff measure) Let $S \subset \mathbb{R}^n$ be a set, for any $d \geq 0$, we define the d -dimensional Hausdorff measure as:

$$\mathcal{H}^d(S) = \liminf_{\delta \rightarrow 0} \left\{ \sum_{i \in I} (\text{diam } U_i)^d \mid S \subset \bigcup_{i=1}^{\infty} U_i, \text{diam } U_i < \delta \right\},$$

where $\{U_i\}_{i=1}^{\infty}$ is a countable cover of S .

The Hausdorff measure just differs by a constant from the Lebesgue measure; $\text{Vol}^d(U) = C_d \mathcal{H}^d(U)$ where C_d depends on d .

Theorem 6.5 (Basic area estimate). *If an algebraic hypersurface S bisects a unit ball, then S has $(n-1)$ -dimensional volume at least $C(n)$.*

Proof. Let $K = B_1$ be the unit ball and $\omega_n = \text{Vol}^n(B_1)$. The hypersurface S bisects K into K_1 and K_2 , so that $\text{Vol}^n(K_1) = \text{Vol}^n(K_2) = \omega_n/2$. Define $K_i^t = K_i + tB_1$, for $i = 1, 2$, with $t > 0$.

Let $n \geq 1$ and μ denote the Lebesgue measure on $\mathbb{R}^n$. Recall the Brunn-Minkowski theorem. It states that if A and B are two compact nonempty subsets of $\mathbb{R}^n$, then the following inequality holds:

$$[\mu(A + B)]^{1/n} \geq [\mu(A)]^{1/n} + [\mu(B)]^{1/n}.$$

Applying this, we see that $[\text{Vol}^n(K_i^t)]^{1/n} \geq [\text{Vol}^n(K_i)]^{1/n} + [\text{Vol}^n(tB_1)]^{1/n} = (\omega_n/2)^{1/n} + (t\omega_n)^{1/n}$. Then

$$\text{Vol}^n(K_1^t) + \text{Vol}^n(K_2^t) \geq 2 \left(\left(\frac{\omega_n}{2} \right)^{1/n} + t\omega_n^{1/n} \right)^n.$$

Notice

$$\begin{aligned} \text{Vol}^n(K_1^t) + \text{Vol}^n(K_2^t) &= \text{Vol}^n(K_1^t \cup K_2^t) + \text{Vol}^n(K_1^t \cap K_2^t) \\ &\leq (1+t)^n \omega_n + \text{Vol}^n(K_1^t \cap K_2^t). \end{aligned}$$

Also, we can show the following (Lemma 6.6):

$$\text{Vol}^n(K_1^t \cap K_2^t) = 2t\mathcal{H}^{n-1}(S) + o(t).$$

Combining the equations above, we get

$$\omega_n[(1 + 2^{\frac{1}{n}}t)^n - 1] \leq \omega_n[(1+t)^n - 1] + 2t\mathcal{H}^{n-1}(S) + o(t).$$

The constant -1 is for us to expand the exponents into series. By Taylor expansion, we simplify the above to

$$\omega_n[n2^{\frac{1}{n}}t + o(t)] \leq \omega_n[nt + o(t)] + 2t\mathcal{H}^{n-1}(S) + o(t),$$

which is,

$$\mathcal{H}^{n-1}(S) \geq \frac{n}{2}\omega_n(2^{\frac{1}{n}} - 1) + o(t) - \frac{o(t)}{2t}.$$

Letting $t \rightarrow 0$, we get

$$\mathcal{H}^{n-1}(S) \geq \omega_n(2^{\frac{1}{n}} - 1) \geq 0,$$

where

$$C(n) = \omega_n(2^{\frac{1}{n}} - 1).$$

□

Lemma 6.6. *Define the hypersurface S and sets K_1^t, K_2^t as above, then we have $\text{Vol}^n(K_1^t \cap K_2^t) = 2t\mathcal{H}^{n-1}(S) + o(t)$.*

We leave the proof of Lemma 6.6 to Appendix A. Finally, we begin the proof of Theorem 1.5. This is an explained and adapted version of the proof from [6].

Proof. The general idea of this proof is as follows. We cover $\mathbb{R}^n$ with unit balls and look at all unit balls that overlap with the set I . Say there are V such unit balls and denote them as $Q_1, \dots, Q_V$. We attempt to find a bound on V , the number of unit balls. Doing so will show our inequality, since $I \subset \{Q_k \mid k \in \{1, \dots, V\}\}$ implies $\text{Vol}(I) \leq V\omega_n$. We know that we have an algebraic hypersurface of degree $d \lesssim_n V^{1/n}$ that bisects all of these unit balls. We call this surface Z . For each ball, it must have at least one tube from every direction going through it. This is because every point $x \in I \subset \mathbb{R}^n$ belongs to at least one tube from every direction, since I is defined to be the intersection of tubes in every direction. Fix a unit ball Q_k . Consider the projection of the area (directed volume) of the algebraic hypersurface Z in every direction (given by cores of tubes) in ball Q_k . We note, by Lemma 5.4, that the sum of these projected volumes is $\gtrsim$ than the $(n-1)$ -dimensional volume of $Q_k \cap Z$, where Q_k is the k^{th} unit ball and Z is our algebraic hypersurface. We can use this lemma since we have at least n vectors in different directions, and the sum of additional vectors is positive anyway, since the directed volume is positive. But we know by Theorem 6.5 that $\text{Vol}(Q_k \cap Z) \gtrsim_n 1$. So we have

$$1 \lesssim_n \text{Vol}(Q_k \cap Z) \lesssim_n \sum_{\substack{v : v \text{ is a core of tube} \\ \text{that intersects } Q_k}} V_{Q_k \cap Z}(v). \quad (8)$$

Therefore, there exists some direction v_k corresponding to a ball Q_k where the projected area is bounded below by a fixed constant that does not vary with V or A . We would like to consider disjoint balls. However, the current family $\mathcal{F} = \{Q_1, \dots, Q_V\}$ is not necessarily disjoint.

We apply the Besicovitch covering lemma, which says $\mathcal{F} = \mathcal{G}_1 \cup \mathcal{G}_2 \cup \dots \cup \mathcal{G}_{N(n)}$ where the unit balls in each family $\mathcal{G}_k$ are pairwise disjoint and the families $\mathcal{G}_k$ partition $\mathcal{F}$. Pick the largest such family, say, $\mathcal{G}' = \{Q'_1, Q'_2, \dots, Q'_{V'}\}$. We have $V' \geq V/N(n)$, where $N(n)$ is the Besicovitch overlap constant.

Now consider the tube with core in the direction of v_k and associate with it the unit ball Q'_k . Do the same with all other balls. Informally, we know that there are many more balls than tubes (if not, the inequality we are trying to show is trivial), thus we use the pigeonhole principle and we find a tube that goes through $\gtrsim V'/A$ balls. We enlarge the radius of this tube to collect all unit balls that intersect it and call it a big tube T . Now for this big tube we have the inequality below, which follows from Lemma 5.3, the inequality (8), the pigeonhole principle as above and noting $V_A(v) + V_B(v) = V_{A \cup B}(v)$. For A and B disjoint,

$$\frac{V}{A} \lesssim_n \frac{V'}{A} \lesssim_n \sum_{\# \text{ unit balls intersecting } T} V_{Q_k \cap Z}(v) \leq V_{Z \cap T}(v) \lesssim d \lesssim V^{1/n},$$

where T is the large tube and v is the core of T . Upon rearranging this inequality, we get

$$\text{Vol}(I) \leq V \omega_n \lesssim A^{\frac{n}{n-1}}$$

as required.

How do we know that V (the number of unit balls covering I) is finite? We give a formal argument below:

Divide $\mathbb{R}^n$ into unit balls. Let $Q_1, Q_2, \dots, Q_V$ be all the balls that intersect with I . Claim that V is finite.

We aim to prove that I is compact. Firstly, it is obvious that I is closed because all the cylinders are closed. Finite unions and intersections of closed sets are closed. We need to prove that I is bounded. We can change the order of the union and the intersection.

$$I = \bigcup_{(a_1, \dots, a_n) \in \{1, \dots, A\}^n} \left(\bigcap_{j=1}^n T_{j, a_j} \right),$$

so we only need to consider the last part of the equation. Let $v_1, v_2, \dots, v_n$ be the unit vectors of the cores of cylinders $T_{1, a_1}, \dots, T_{n, a_n}$. We show that these vectors are linearly independent.

Consider the $n \times n$ matrix $V = (v_1, \dots, v_n)$. We need to show that it is invertible. Let $\theta = (100n)^{-1}$. We know $|v_i \cdot e_i| \geq \cos(\theta)$ and $|v_i \cdot e_j| \leq \sin(\theta)$. Let σ_i be the eigenvalues of VV^* . We know $\sigma_{\min}(A) \geq \sigma_{\min}(B) - \|A - B\|_2$. We consider the matrix $V - I$. For diagonal entries, $|(V - I)_{ii}| = |v_i \cdot e_i - 1| \leq 1 - \cos(\theta) \leq \frac{\theta^2}{2}$.

We can bound $\|V - I\|_2$ by the Frobenius norm $\|V - I\|_F$. We have

$$\|V - I\|_F = \sqrt{\sum_{i=1}^n \sum_{j=1}^n |(V - I)_{ij}|^2} \leq \sqrt{\frac{n\theta^4}{4} + n(n-1)\theta^2} \leq \sqrt{n\theta^2 + n(n-1)\theta^2} = n\theta.$$

Hence, $\|V - I\|_2 \leq \|V - I\|_F \leq n\theta = \frac{1}{100}$. So $\sigma_{\min}(V) \geq 1 - \|V - I\|_2 \geq \frac{99}{100}$ and all the eigenvalues of VV^* are greater than 0. We get that V is invertible, which means the v_i 's are linearly independent.

For each core L_j , the cylinders j are the set of points satisfying $\text{dist}(x, L_j) \leq 1$. It can be written as $\|\pi_{v_j}(x - p_j)\| \leq 1$, where $\pi_{v_j}(x) : \mathbb{R}^n \rightarrow v_j^\perp$ is the orthogonal projection to $v_j^\perp$ and p_j is an arbitrary point in the core L_j . Let A_j be the matrix such that $A_j(x) = \pi_{v_j}(x)$. We have $A_j = I - v_j v_j^T$. The set $S = \bigcap_{j=1}^n T_{j, a_j} = \bigcap_{j=1}^n \{x : \|\pi_{v_j}(x - p_j)\| \leq 1\}$.

We can show that S is bounded now. Suppose S is unbounded. Then there exists a sequence x_k in S such that $\|x_k\| \rightarrow \infty$. Let $y_k = \frac{x_k}{\|x_k\|}$. For any $1 \leq j \leq n$, we have $\|\pi_{v_j}(x_k - p_j)\| = \|A_j(x_k - p_j)\| \leq 1$, which implies $\left\|A_j(y_k - \frac{p_j}{\|x_k\|})\right\| \leq \frac{1}{\|x_k\|} \rightarrow 0$. Since $\frac{A_j p_j}{\|x_k\|} \rightarrow 0$ as $k \rightarrow \infty$, we get $\|A_j y_k\| \rightarrow 0$. Since $\|y_k\| = 1$, by Bolzano-Weierstrass, there is a subsequence of y_k that converges to y . We have $\|y\| = 1$ and $\|A_j y\| = 0$ for all j . However, $\|A_j y\| = 0$ implies y is parallel to v_j . So y is parallel to all v_j for $1 \leq j \leq n$. But $(v_1, v_2, \dots, v_n)$ are linearly independent, so they make up a basis of $\mathbb{R}^n$, which means $y = 0$ is the only solution. Contradiction. S is bounded.

A finite union of bounded sets is also bounded, so I is bounded. I is closed and bounded implies I is compact. So we have V is finite. $\square$

7 Further results

7.1 Endpoint case of the multilinear Kakeya estimate

Theorem 1.5 can be taken further. In [6], Guth proved the following more general result.

Theorem 7.1 (Endpoint case of the multilinear Kakeya estimate, [6]). *Let $T_{j,a}$ be arbitrary cylinders in $\mathbb{R}^n$ with radius 1 and infinite length, ranging over $1 \leq j \leq n$ and $1 \leq a \leq A(j)$. Let $v_{j,a}$ be a unit vector parallel to $T_{j,a}$. If the determinant of the matrix $(v_{1,a_1}, \dots, v_{n,a_n})$ has a norm of at least $\theta > 0$ where $a_j \in \{1, \dots, A(j)\}$, then*

$$\int \prod_{j=1}^n \left(\sum_{a=1}^{A(j)} 1_{T_{j,a}} \right)^{\frac{1}{n-1}} \leq C_{n,\theta} \prod_{j=1}^n A(j)^{\frac{1}{n-1}}.$$

As Guth points out in [6], there are several ways in which Theorem 7.1 is stronger than Theorem 1.5.

First, while Theorem 1.5 imposes a restriction on the cylinders to be nearly parallel to the axes, Theorem 7.1 relaxes this assumption, imposing a gentler transversality hypothesis and it can be shown that our assumptions on the directions of cylinders in weaker version of the theorem satisfy the transversality hypothesis.

Second, Theorem 7.1 allows for the number of cylinders in each class j to vary depending on j , whereas this number is a constant A in Theorem 1.5.

Third, the bound in Theorem 7.1 is more meaningful, because the integrand is a function which depends on the number of overlapping tubes from different classes (i.e. it counts multiplicity in the case where a point is contained within more than one tube in the same family). In contrast, Theorem 1.5 is just a bound on the support of this function.

7.2 Extension from the endpoint case

Although Bennett, Carbery and Tao proved Theorem 1.4 for $\frac{n}{n-1} < q \leq \infty$ using the heat flow method, we observe here that Theorem 7.1 alone can be used to imply the multilinear Kakeya estimate for all $\frac{n}{n-1} \leq q < \infty$. We will make use of the following theorem:

Theorem 7.2 (Hadamard three-line theorem). *Let f be a complex function defined on the strip $S = \{x + iy : a \leq x \leq b, y \in \mathbb{R}\}$ bounded and continuous on S and holomorphic in the interior of S . If $M(x) = \sup_{y \in \mathbb{R}} |f(x + iy)|$, then*

$$M(x) \leq M(a)^{(b-x)/(b-a)} M(b)^{(x-a)/(b-a)} \quad \text{for all } a \leq x \leq b,$$

i.e. $\log M(x)$ is convex on $[a, b]$.

Now recall from (4) that the endpoint case of the multilinear Kakeya estimate can be expressed as

$$\int_{\mathbb{R}^n} \prod_{j=1}^n \left(\sum_{a=1}^{A(j)} 1_{T_{j,a}} \right)^{\frac{1}{n-1}} \leq C_{n,\theta} \prod_{j=1}^n \delta A(j)^{\frac{1}{n-1}}$$

when $T_{j,a}$ are cylinders of radius $\delta \in (0, 1)$ and length 1. Following from the above inequality, let

$$F(z) = \int_{\mathbb{R}^n} \prod_{j=1}^n \left(\sum_{a=1}^{A(j)} 1_{T_{j,a}} \right)^z, \quad G(z) = \prod_{j=1}^n \delta A(j)^z,$$

and $f(z) = F(z)/G(z)$ be functions defined on the complex plane $\mathbb{C}$. Our aim is to show that $f(z)$ is uniformly bounded (over all possible choices of δ and cylinders $T_{j,a}$) on $S = \{z \mid \operatorname{Re}(z) \geq \frac{1}{n-1}\}$.

First, we have that

$$|f(x + iy)| = \left| \frac{F(x + iy)}{G(x + iy)} \right| \leq \left| \frac{F(x)}{G(x)} \right| = |f(x)| \quad \text{for all } x > 0, y \in \mathbb{R},$$

since

$$|F(x + iy)| \leq \int_{\mathbb{R}^n} \left| \prod_{j=1}^n \left(\sum_{a=1}^{A(j)} 1_{T_{j,a}} \right)^{x+iy} \right| = \int_{\mathbb{R}^n} \prod_{j=1}^n \left(\sum_{a=1}^{A(j)} 1_{T_{j,a}} \right)^x = F(x)$$

and

$$|G(x + iy)| = \prod_{j=1}^n \delta |A(j)^{x+iy}| = \prod_{j=1}^n \delta A(j)^x = G(x).$$

Hence, Theorem 7.1 implies that $|f(\frac{1}{n-1} + iy)| \leq C_{n,\theta}$ for all $y \in \mathbb{R}$. Also, when $R > \frac{1}{n-1}$ is a sufficiently large constant, $|f(R + iy)| \leq C'_n \delta^{-n} 2^R$ for all $y \in \mathbb{R}$, where C'_n is some constant. This follows from

$$F(R) \leq \left(\prod_{j=1}^n A(j) \right)^R \operatorname{Vol} \left(\bigcap_{j=1}^n \bigcup_{a=1}^{A(j)} T_{j,a} \right) = \delta^{-n} G(R) \operatorname{Vol} \left(\bigcap_{j=1}^n \bigcup_{a=1}^{A(j)} T_{j,a} \right).$$

Now consider the region $S_R = \{z \mid \frac{1}{n-1} \leq \operatorname{Re}(z) \leq R\}$. Since f is bounded and continuous on S_R and holomorphic in the interior of S_R , by Theorem 7.2,

$$|f(p)| \leq C_{n,\theta}^{(R-p)/(R-\frac{1}{n-1})} (C'_n \delta^{-n} 2^R)^{(p-\frac{1}{n-1})/(R-\frac{1}{n-1})} \quad \text{for all } \frac{1}{n-1} \leq p \leq R.$$

Taking $R \rightarrow \infty$ on the right hand side, we obtain the limit $2^{p-\frac{1}{n-1}} C_{n,\theta}$. Thus,

$$|f(p)| \leq 2^{p-\frac{1}{n-1}} C_{n,\theta} \quad \text{for all } p \geq \frac{1}{n-1}. \quad (9)$$

Since the bound $2^{p-\frac{1}{n-1}}C_{n,\theta}$ is independent of δ and the cylinders $T_{j,a}$, we have finished proof that $f(z)$ is uniformly bounded on S . Hence, we have the multilinear Keakeya estimate for cases $\frac{n}{n-1} \leq q < \infty$.

The case $q = \infty$ is by itself straightforward. The inequality

$$\left\| \prod_{j=1}^n \left(\sum_{a=1}^{A(j)} 1_{T_{j,a}} \right) \right\|_{L^\infty(\mathbb{R}^n)} \leq C_{n,\theta} \prod_{j=1}^n A(j)$$

would always hold since the L^∞ -norm can not be greater than the product of all $A(j)$. Thus, we see that the endpoint case is sufficient to view all other cases of the multilinear Keakeya estimates.

A Appendix: Proof of Lemma 6.6

Proof of Lemma 6.6. Let $d(x) = \text{dist}(x, S)$. Denote the set $N_t(S) = \{x \in \mathbb{R}^n \mid |d(x)| \leq t\}$. By the definition of K_1^t and K_2^t , it is obvious that $N_t(S) = K_1^t \cap K_2^t$. We only prove for S being $C^{1,1}$. There exists a small ρ such that for every x with $d(x) \leq \rho$, there is a unique nearest point on S . Define $\Phi : S \times (-\rho, \rho) \rightarrow \mathbb{R}^n$ such that $\Phi(p, s) = p + s\nu(p)$ where $\nu(p)$ denotes the unit normal at point p .

By Coarea formula, for u a real-valued Lipschitz function on Ω and $g \in L^1$,

$$\int_{\Omega} g(x) \|\nabla u(x)\| dx = \int_{\mathbb{R}} \int_{u^{-1}(t)} g(x) d\mathcal{H}^{n-1}(x) dt$$

Take $u(x) = d(x)$, $\Omega = N_t(S)$ $g(x) = 1$, since $\|\nabla d(x)\| = 1$, we compute

$$\begin{aligned} \text{LHS} &= \int_{N_t(S)} dx = \text{Vol}^n(N_t(S)) \\ \text{RHS} &= \int_{-\infty}^{\infty} \mathcal{H}^{n-1}(\{x : d(x) = s\}) ds = \int_{-t}^t \mathcal{H}^{n-1}(\{x : d(x) = s\}) ds \end{aligned}$$

We get

$$\text{Vol}^n(N_t(S)) = \int_{-t}^t \mathcal{H}^{n-1}(\{x : d(x) = s\}) ds$$

For $|s| \leq \rho$, $\Phi_s(p) = p + s\nu(p)$. $T_p S = \{\gamma'(0) : \gamma : (-\epsilon, \epsilon) \rightarrow S, \gamma(0) = p\}$. Pick arbitrary $v \in T_p S$, then

$$\left. \frac{d}{dt} \right|_{t=0} \Phi_s(\gamma(t)) = \left. \frac{d}{dt} \right|_{t=0} \gamma(t) + s\nu(\gamma(t)) = v + sD\nu(p)[v]$$

Since $\nu(p)$ is the unit normal, we have $\langle \nu(x), \nu(x) \rangle = 1$. Differentiate the equation in direction $v \in T_p S$, $\left. \frac{d}{dt} \right|_{t=0} \langle \nu(\gamma(t)), \nu(\gamma(t)) \rangle = 2\langle \nu(p), D\nu(p)[v] \rangle$, by the other definition of $T_p S$, we have $D\nu(p)[v] \in T_p S$. Define the Weingarten map $W_p : T_p S \rightarrow T_p S$ as

$$W_p(v) = -D\nu(p)[v].$$

So we have

$$D\Phi_s(p)[e_i] = (I + sW_p)e_i.$$

At any point p , we have an orthonormal basis $e_1, \dots, e_{n-1}$ that diagonalize W_p

$$W_p e_i = \kappa_i e_i$$

for $i = 1, \dots, n-1$. So we can calculate the determinant

$$\det(D\Phi_s(p)) = \prod_{i=1}^{n-1} (1 + s\kappa_i) = 1 + s \sum_{i=1}^{n-1} \kappa_i + o(s^2).$$

Let $\kappa = \sum_{i=1}^{n-1} \kappa_i$. According to change of variables,

$$\begin{aligned} \mathcal{H}^{n-1}(\{x : d(x) = s\}) &= \int_S |\det(D\Phi_s(p))| d\mathcal{H}^{n-1}(p) \\ &= \int_S (1 + s\kappa + o(s^2)) d\mathcal{H}^{n-1} \\ &= (1 + o(s^2)) \mathcal{H}^{n-1}(S). \end{aligned}$$

Combine it with the equation above,

$$\text{Vol}^n(N_t(S)) = \int_{-t}^t (1 + o(s^2)) \mathcal{H}^{n-1}(S) ds = 2t \mathcal{H}^{n-1}(S) + o(t^3).$$

□

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